Harness

File: harness.kicad_sch

Daughterboard

File: daughter.kicad_sch

LSU, MAP, IAT, USB, AUX

File: io.kicad_sch

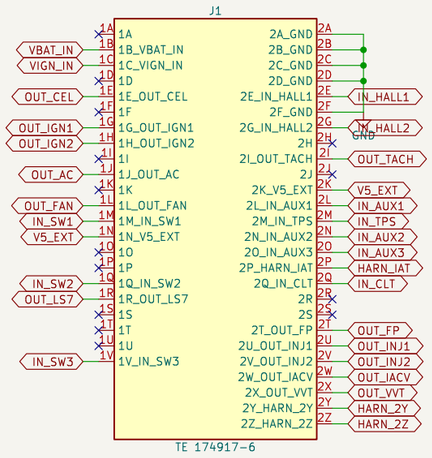
Miata ECU motherboard 1989–1993 (NA6). Carries the year-specific harness header and the year-independent I/O (wideband, MAP, IAT, USB-C). All engine control is on the daughterboard.

Sheet: /
File: miata_mb_8993.kicad_sch

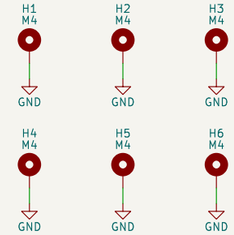
Title: Miata ECU motherboard 1989–1993 (NA6)

Size: A4	Date: 2026-10-30	**Rev: 0.10**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/4

Stock harness header (Miata ECU motherboard 1989–1993): every pin labelled with the daughterboard signal it drives



Case mounting bosses (M4), tied to ground



Generated from scripts/mb_na6.py; netlist identical to the routed board

Sheet: /Harness/
File: harness.kicad_sch

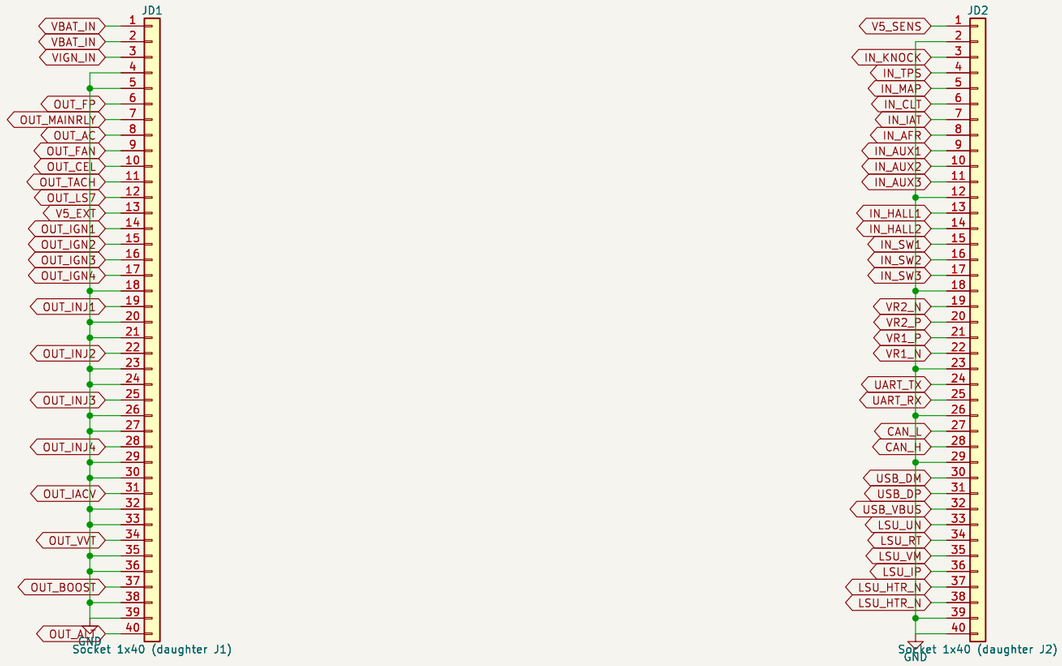
Title: Miata ECU motherboard 1989–1993 — Harness

Size: A3	Date: 2026-10-30	**Rev: 0.10**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 2/4

Daughterboard sockets: pin-for-pin with the daughterboard J1 (top) and J2 (bottom)

Daughterboard standoffs (M3)

HD1 M3 standoff
HD2 M3 standoff
HD3 M3 standoff
HD4 M3 standoff



Generated from scripts/mb_na6.py; netlist identical to the routed board

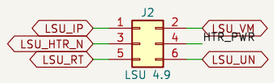
Sheet: /Daughterboard/
File: daughter.kicad_sch

Title: Miata ECU motherboard 1989–1993 — Daughterboard

Size: A3	Date: 2026-10-30	**Rev: 0.10**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/4

LSU 4.9 wideband sensor (controller on the daughterboard)

LSU 4.9: 1 IP 2 VM 3 H− 4 H+ 5 RT 6 UN

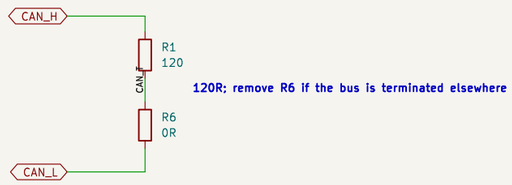




Heater fuse: 2 A hold, 30 V PTC

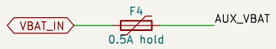
IP/VM/RT/UN and H− run to daughterboard J2 33–38; the heater current returns on J2 39/40

CAN bus termination at the ECU end

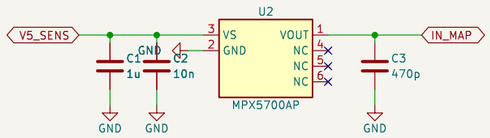


120R; remove R6 if the bus is terminated elsewhere

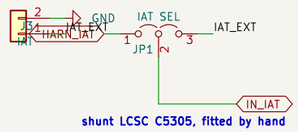
Fused +12 V feeds (harness actuators, AUX)



MAP sensor

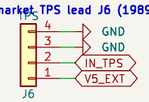


IAT source jumper JP1: 2–3 EXT = onboard connector J3 (default), 1–2 STOCK = stock harness sensor

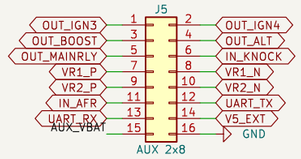


shunt LCSC C5305, fitted by hand

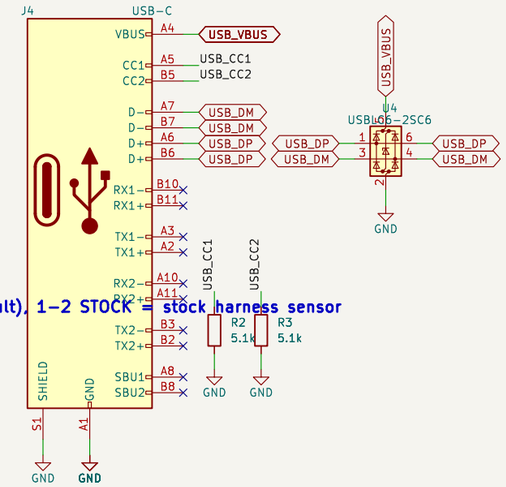
Aftermarket TPS lead J6 (1989–93 manual cars): pin 2 is the same input as harness 2M



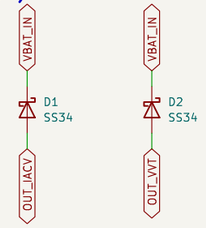
AUX: add−on I/O the stock harness has no pins for (+12 V through F4)



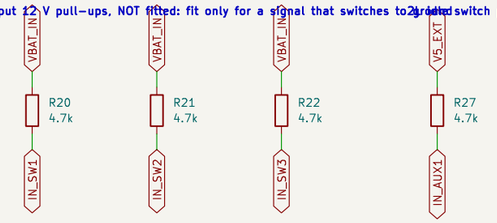
USB−C (device): tuning, logs, DFU firmware update



Valve freewheel diodes (to each valve's +12 V feed) and fit options

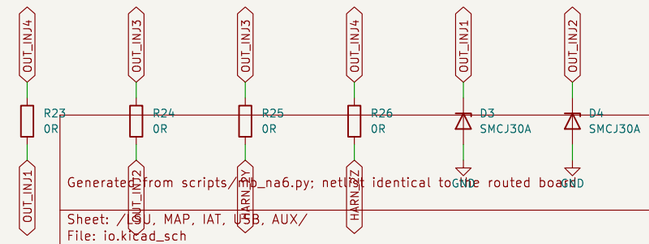


Switch−input 12 V pull−ups, NOT fitted: fit only for a signal that switches to ground; add a switch pull−up (NOT fitted)



1989–93 batch harness: two drivers per injector wire, TVS clamp per wire

Sequential conversion: remove R23/R24, fit R25/R26 (never R26 on an automatic)



Generated from scripts/build_na6.py; netlist identical to the routed board

Sheet: /GPIO, MAP, IAT, USB, AUX/
File: io.kicad_sch

Title: Miata ECU motherboard 1989–1993 – LSU, MAP, IAT, USB, AUX

Size: A3	Date: 2026−10−30	**Rev: 0.10**
KiCad E.D.A. kicad−cli 7.0.11+dfsg−1build4		Id: 4/4